

Exhibit 11: Cash Flows for a Dividend-Paying Stock

Time	Outflows
0	USD200 to purchase the first share
1	USD225 to purchase the second share
Time	Inflows
1	USD5 dividend received from first share (and not reinvested)
2	USD10 dividend (USD5 per share × 2 shares) received
2	USD470 received from selling two shares at USD235 per share

To solve for the money-weighted return, the first step is to group net cash flows by time. For this example, we have –USD200 for the $t = 0$ net cash flow, –USD220 = –USD225 + USD5 for the $t = 1$ net cash flow, and USD480 for the $t = 2$ net cash flow. After entering these cash flows, we use the spreadsheet's (such as Excel) or calculator's IRR function to find that the money-weighted rate of return is 9.39 percent.

$$\begin{aligned}
 CF_0 &= -200 \\
 CF_1 &= -220 \\
 CF_2 &= +480 \\
 \frac{CF_0}{(1 + IRR)^0} + \frac{CF_1}{(1 + IRR)^1} + \frac{CF_2}{(1 + IRR)^2} &= 0 \\
 \frac{-200}{1} + \frac{-220}{(1 + IRR)^1} + \frac{480}{(1 + IRR)^2} &= 0 \\
 IRR &= 9.39\%
 \end{aligned}$$

All these calculations can be performed using Excel using the =IRR(values) function, which results in an IRR of 9.39 percent.

Now we take a closer look at what has happened to the portfolio during each of the two years.

In the first year, the portfolio generated a one-period holding period return of $(USD5 + USD225 - USD200)/USD200 = 15\%$. At the beginning of the second year, the amount invested is USD450, calculated as USD225 (per share price of stock) × 2 shares, because the USD5 dividend was spent rather than reinvested.

At the end of the second year, the proceeds from the liquidation of the portfolio are USD470 plus USD10 in dividends (as outlined in Exhibit 11). So, in the second year the portfolio produced a holding period return of $(USD10 + USD470 - USD450)/USD450 = 6.67\%$. The mean holding period return was $(15\% + 6.67\%)/2 = 10.84\%$.

The money-weighted rate of return, which we calculated as 9.39 percent, puts a greater weight on the second year's relatively poor performance (6.67 percent) than the first year's relatively good performance (15 percent), as more money was invested in the second year than in the first. That is the sense in which returns in this method of calculating performance are "money weighted."

Although the money-weighted return is an accurate measure of what the investor earned on the money invested, it is limited in its applicability to other situations. For example, it does not allow for a return comparison between different individuals or different investment opportunities. Importantly, two investors in the *same* mutual fund or with the same portfolio of underlying investments may have different money-weighted returns because they invested different amounts in different years.